

Name: _____

Date: _____

Class: _____

MyDesign Portfolio

Element D Initial Template AT - 15 points

Guidance before filling out the template on the next page:

Because you will iterate (repeat) your design multiple times, you will need to fill out multiple copies of these sections. Think of Element D as a diary that you will date and keep adding to throughout your work until your final iteration. Iteration must shine through in this Element.

You should begin with many design ideas that reflect the individual ideas of your group members and then the group ideas. Number these ideas and include a date, description, drawing, and details for each.

After you have created an initial set of ideas, document your decision-making on which design to include with a design matrix. Create a plan of work, ideally using a Gantt chart that may be housed in this document or in a different day-to-day document. Generate a design blueprint with enough details that someone else not familiar with your project could also build your design.

You will then proceed forward to build and test this design, returning to Element D afterward when you are ready to create more design ideas. Fill out each of these steps again.

Element D: Design Concept Generation, Analysis, and Selection

Design Idea Generation (D1)

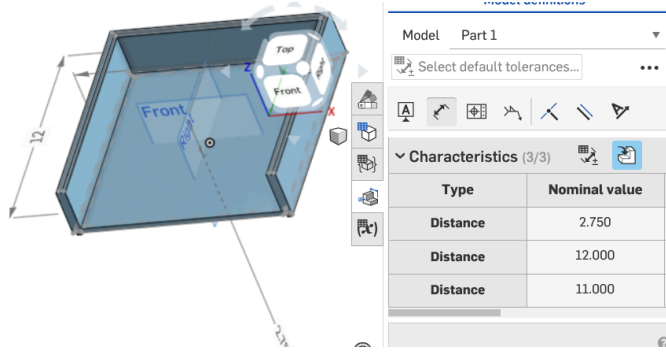
This section focuses on the process of generating solutions. The initial rounds should include every idea you come up with but the subsequent rounds of ideation should lead to viable options. Each idea should include not only words, but also a picture. Use as many copies of this sub-template as needed.

Name: _____

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Design Idea #1	Drawers and Chromebook Rack.
Date:	2026-03-06T07:50
Description:	One stack of drawers used to store miscelanius materials connected to a second stack used for the storage and charging of chromebooks.
Drawing that includes distinguishing information such as scale, dimensions, and materials:	 metersSurrattDrawerInitialSketch2026-03-06T08:01.pdf

Design Idea #2	drawers								
Date:	2026-03-06T08:05								
Description:	Stackable drawers that can store stuff								
Drawing that includes distinguishing information such as scale, dimensions, and materials:	 The image shows a 3D model of a stackable drawer. The model is blue and has dimensions of 12, 12, and 12. It is labeled 'Front' and 'Top'. To the right of the model is a software interface with a table of characteristics. <table border="1"> <thead> <tr> <th>Type</th> <th>Nominal value</th> </tr> </thead> <tbody> <tr> <td>Distance</td> <td>2.750</td> </tr> <tr> <td>Distance</td> <td>12.000</td> </tr> <tr> <td>Distance</td> <td>11.000</td> </tr> </tbody> </table>	Type	Nominal value	Distance	2.750	Distance	12.000	Distance	11.000
Type	Nominal value								
Distance	2.750								
Distance	12.000								
Distance	11.000								

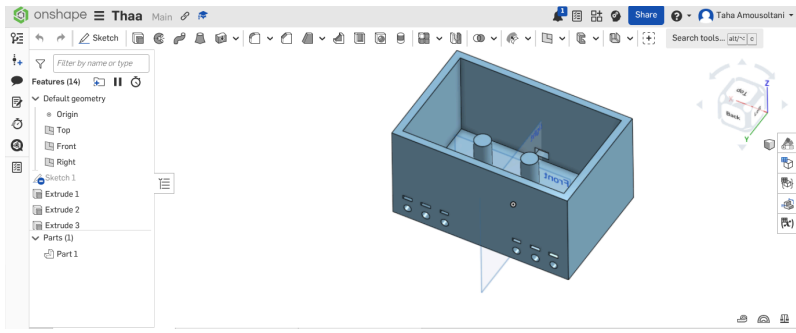


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Class: _____

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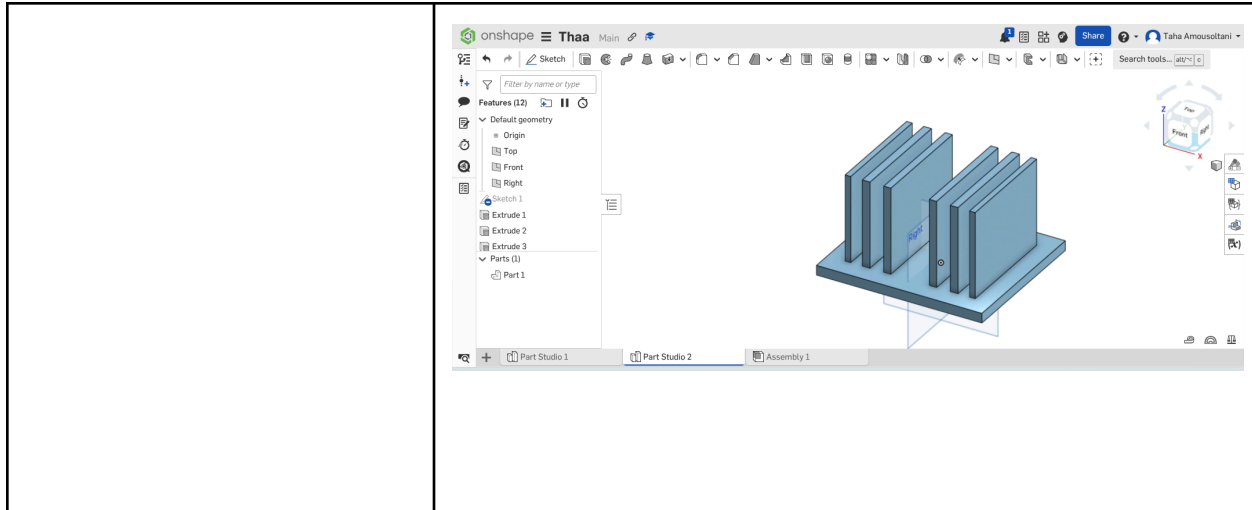
Design Idea #3	Chrome book Charging station (Taha)
Date:	3/6/2026
Description:	<i>It is a Box with a hole for the chromebook charger to plug in to wall and poles to wrap the wires to organize the wires and holes in the front that allow the usb C charger to pass through to the chromebook slots on the top that hold the chromebook for secure charging , it will be printed in two parts with PLA or ABS for a more premium durable material.</i>
Drawing that includes distinguishing information such as scale, dimensions, and materials:	



Name: _____

Date: _____

Class: _____



Design Idea #4	
Date:	
Description:	
Drawing that includes distinguishing information such as scale, dimensions, and materials:	



Name: _____

Date: _____

Class: _____

Design Idea #5 (if needed)	
Date:	
Description:	
Drawing that includes distinguishing information such as scale, dimensions, and materials:	

Design Idea Comparison and Selection (D1, D2)

When you've finished generating ideas, you must compare your design options according to all your design requirements. Incorporate the priority of those requirements when comparing design options.

A decision matrix is used to allow for systematic, weighted comparisons.

If using weights in a decision matrix, be sure to explain and justify the weighting based on your design requirements.

You may use the weighted design matrix template below, inserting your criteria and design options.



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Design Requirement	Weight (multiplied by each design score)(Design 1)	Nick's Drawer Design		Meter's Modular Multipurpose Design(Design 2)		Taha's 'Rack' "Design" (Design 3)	
Ease of Access	5	5	25	5	25	4	21
Durability	3	4	12	5	15	3	11
Storage Capacity	4	4	16	5	20	3	14
Cost	3	3	9	2	6	2	6
Ease of Construction	2	5	10	4	8	2	7
Score Unweighted	N/a	21		21		14	
Total score	N/a	72		74		59	

Our team chooses design 2 to solve the chromebook charging problem because our primary stakeholder, Mr HS needs a solution to his chromebook chargers being disorganized and stolen. (which specific requirements does it fill).

Design Blueprint (D4)

Describe all of your design details so that someone unfamiliar with your project can build your prototype.

Include orthographic sketches (front, top, and side views), dimensions (include units), CAD drawings (if taught in your class).

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What manufacturing methods, material choices, supply lists, and costs do you anticipate for the prototype?

When you iterate on this idea, you will copy the design blueprint section for the second design.

Design Blueprint #1

Date:

Sketch or CAD Drawing (orthographic/multiview):

Manufacturing Methods (cut, saw, paste, screw, tie, glue):

Step 1: FDM(3D) print components.

Step 2: After Printing assemble components.

Step 3: After assembly install chromebook chargers.

Supply List and Costs:

Item	Cost each	# needed	Total
PLA	20\$/kg	300 grams	\$6
Zipties	0.5\$/per unit	3 units	0.15\$
Total Cost			\$6.15

Plan of Action (D3)

- A Gantt chart is encouraged to outline the primary steps expected on a visual timeline. [Here is one you can use.](#)
- Larger tasks from the Gantt chart should be broken down further into detailed sub-tasks, which may be housed in a different day-to-day document.
- The Gantt chart will be a future assignment so if you start now you will be ahead.
- Discrete tasks should include enough detail such that others could replicate the efforts, such as procedures, materials, tools, status, etc.
- Make screenshots of the document instead of using the link.



Portfolio - Element D Initial Template

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Wait until you create and test your first prototype to fill out the next section.

Iterations for the Solution (How did you change your prototype after building and testing it?):

Date of Iteration:

Description of Iteration:

Images if applicable:

Date of Iteration:

Description of Iteration:

Images if applicable:

Date of Iteration:

Description of Iteration:

Images if applicable:

Rubric

Element D: Design Concept Generation,						
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Name: _____

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Analysis and Selection						
	5	4	3	2	1	0
1. Process for Generating & Comparing Possible Solutions. The Process for Generating Possible Solutions _____.	<i>was comprehensive, iterative, and consistently defensible (viable design highly likely)</i>	<i>was thorough, iterative, and generally defensible (viable design likely)</i>	<i>was adequate, generally iterative and defensible (viable design is possible)</i>	<i>was partial or overly general and only somewhat iterative and/or defensible (raising issues with the design viability)</i>	<i>was incomplete or minimally iterative and/or defensible (raising issues with the design viability)</i>	<i>was not evident that there was an attempt</i>
2. Design Solution Choice. The Design Solution chosen _____.	<i>was well-justified and demonstrated attention to all design requirements</i>	<i>was justified and demonstrated attention to most if not all, design requirements</i>	<i>was explained with reference to at least some design requirements</i>	<i>was not sufficiently explained with reference to design requirements</i>	<i>lacked support related to design requirements</i>	<i>was not attempted or evidence not provided</i>
3. Plan of Action. The plan of action _____.	<i>has considerable merit and would easily support repetition and testing for effectiveness by others</i>	<i>would support repetition and testing for effectiveness by others</i>	<i>might not clearly or fully support repetition and testing for effectiveness by others</i>	<i>has insufficient detail to allow for testing for replication of results</i>	<i>is lacking to the point that the design solution cannot be tested</i>	<i>cannot be executed or does not exist</i>

